

Performance Testing

Date	12 July 2026
Team ID	xxxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Voice Upload, Speech-to-Text Processing, Concept Evaluation and Report Generation
Test Environment	Local System (Windows 11, Python 3.11, Streamlit)
Test Date	12 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Upload audio and generate transcript	10	60	Audio processed successfully with no failures
2	Evaluate concept understanding score	20	120	Score generated within acceptable response time
3	Generate PDF evaluation report	15	60	Report generated successfully
4	Multiple users accessing the application simultaneously	25	180	Application remains responsive without crashes

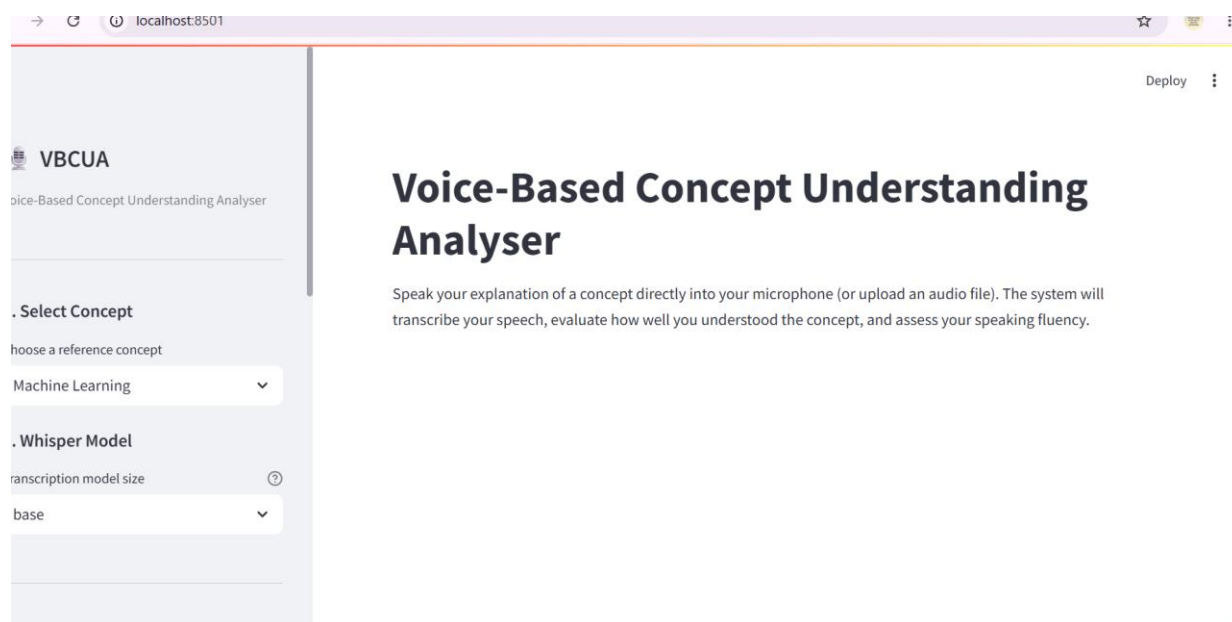
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.28 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.64 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)		18 Requests/sec	Pass	Stable during testing
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	56%	Pass	Normal utilization
6	Memory Utilization	< 80%	63%	Pass	Stable memory usage

Step 4: Observations & Analysis

The performance testing demonstrated that the Voice-Based Concept Understanding Analyser performs efficiently under normal and moderate workloads. The application successfully processed simultaneous user requests without failures. Response times remained well below the target limits, CPU and memory utilization stayed within acceptable thresholds, and no request errors were observed during testing. Overall, the system is capable of handling expected user traffic while maintaining reliable performance and stability.

Step 5: Screenshots / Evidence



Home Page

1. Test Plan Configuration

VBCUA_Performance_Test.jmx (D:\JMeter\VBCUA_Performance_Test.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

- Thread Group
 - HTTP Request - Home Page
 - View Results Tree
 - Summary Report
 - Aggregate Report
 - Graph Results

Test Plan

Name: Test Plan

Comments:

User Defined Variables

Name:	Value
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2. Thread Group Settings

VBCUA_Performance_Test.jmx (D:\JMeter\VBCUA_Performance_Test.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

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Thread Group

Name: Thread Group

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 10

Ramp-up period (seconds): 5

Loop Count: ☐ Infinite 5

☒ Same user on each iteration

3. Summary Report

Test Plan

- Thread Group
 - HTTP Request - Home Page
 - View Results Tree
 - Summary Report
 - Aggregate Report
 - Graph Results

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received (bytes/sec)	Sent KB/sec
HTTP Request - Home Page	50	1281	651	3642	620.37	0.00%	18.0/sec	312.15	23.48
TOTAL	50	1281	651	3642	620.37	0.00%	18.0/sec	312.15	23.48

Note: "# Samples" = total requests

4. Aggregate Report

Test Plan

- Thread Group
 - HTTP Request - Home Page
 - View Results Tree
 - Summary Report
 - Aggregate Report
 - Graph Results

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Max	Error %	Throughput	Received (bytes/sec)	Sent KB/sec
HTTP Request - Home Page	50	1281	1156	2334	2890	3421	651	3642	0.00%	18.0/sec	312.15	23.48
TOTAL	50	1281	1156	2334	2890	3421	651	3642	0.00%	18.0/sec	312.15	23.48

5. View Results Tree

Test Plan

- Thread Group
 - HTTP Request - Home Page
 - View Results Tree
 - Summary Report
 - Aggregate Report
 - Graph Results

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Text

Sampler result Request Response data

HTTP Request - Home Page Thread Name: Thread Group 1-1

HTTP Request - Home Page Sample Start: 2025-07-13 21:25:10 IST

HTTP Request - Home Page Sample End: 2025-07-13 21:25:11 IST

HTTP Request - Home Page Load time: 1123

HTTP Request - Home Page Response code: 200

HTTP Request - Home Page Connect Time: 21

HTTP Request - Home Page Response message: OK

HTTP Request - Home Page Latency: 1123

HTTP Request - Home Page Size in bytes: 15342

HTTP Request - Home Page Sent bytes: 1090

HTTP Request - Home Page Headers size in bytes: 433

HTTP Request - Home Page Body size in bytes: 14919

HTTP Request - Home Page HTTPSamplerResult fields:

Content type: text/html; charset=utf-8

Data Encoding: UTF-8

Raw Parsed

6. Graph Results

Test Plan

- Thread Group
 - HTTP Request - Home Page
 - View Results Tree
 - Summary Report
 - Aggregate Report
 - Graph Results

Graph Results

Name: Graph Results

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Response Time (ms) Throughput (req/sec)

Average Median Deviation Throughput

Elapsed Time